A Proposed MMU core offering backwards compatibility with the 68851.

rf68851

Paged Memory Management Unit

Robert Finch

Goals

- Backwards compatibility.
  - Backwards compatibility is important to leverage existing software base.
- Support for 64-bit virtual and physical addresses.
  - Memory sizes in many cases exceed the address range of 32-bits.
- Support for larger page sizes.
  - As memory systems have grown it is more efficient to use a larger page size, 4kB and larger are popular with contemporary CPUs. For some purposes a MB page size is useful, but the 68851 only supports up to 32kB.
- Support for modern OS.
  - A modern OS may make use of more protection rings than the typical two ring scheme. Providing more protection rings eases the implementation of virtual machines.
- To be innovative rather than inventive.
  - An innovative approach takes an older design an renews it. Often older designs are very good, suited to the times they were created in. Being innovative provides an upgrade path for older designs.
- "Power-on" operation of the MMU.
  - The MMU should be able to translate addresses from a power-on state. Power on operation of the MMU may aid the security and integrity of the system.

Root Pointers

The MMU contains five root pointer registers: the existing supervisor, CPU and DMA registers of the 68851 plus hypervisor and secure operating mode registers. Root pointer registers are extended to allow for up to 64-bit physical addresses.

CRP / ARP	CPU (user/app) root pointer
SRP	Supervisor root pointer
HRP	Hypervisor root pointer
MRP	Secure machine root pointer
DRP	DMA root pointer

Root Pointer

Root pointer register supplies the entry point into the page tables as per the 68851.

<table>
<tr><td></td><td colspan="11">Root Pointer</td><td></td></tr>
<tr><td>63</td><td>L/U</td><td colspan="10">$Limit_{14...0}$</td><td>48</td></tr>
<tr><td>47</td><td>$\sim_3$</td><td>$\sim_3$</td><td>SG</td><td>~</td><td>~</td><td>~</td><td>~</td><td>~</td><td>~</td><td>~</td><td>DT_2</td><td>32</td></tr>
<tr><td>31</td><td colspan="11">$PA_{31...16}$</td><td>16</td></tr>
<tr><td>15</td><td colspan="8">$PA_{15...4}$</td><td>~</td><td>~</td><td>$\sim_2$</td><td>0</td></tr>
</table>

Root Pointer Extension (RPX)

The root pointer extension supplies the upper 32-bits of a 64-bit physical address for the root pointer. It also supplies additional bits for the limit, allowing up to eight million entries in the root table. An additional bit is provided to extend the number of descriptor types to eight.

The default value of this register is zero except for the LVL field which is equal to two.

The default setting provides a high level of compatibility with the 68851 on reset.

<table>
<tr><td></td><td colspan="6">Root Pointer Extension</td><td></td></tr>
<tr><td>63</td><td colspan="6">$PA_{63...48}$</td><td>48</td></tr>
<tr><td>47</td><td colspan="6">$PA_{47...32}$</td><td>32</td></tr>
<tr><td>31</td><td colspan="2">LVL_3</td><td>RGN_3</td><td colspan="2">$\sim_{4...0}$</td><td>$\sim_{4...0}$</td><td>16</td></tr>
<tr><td>15</td><td>DT</td><td colspan="3">$\sim_7$</td><td colspan="2">$Limit_{22...15}$</td><td>0</td></tr>
</table>

Translation Control Registers

There is a separate pair (TC and XTC) of translation control registers for each operating mode. This allows page sizes to be different for each operating mode.

Translation Control Register (TC)

	Translation Control Register						
31	E	0_5	SRE	FCL	PS	IS	16
15	TIA		TIB		TIC	TID	0

Extended Translation Control Register (XTC)

The extended translation control register contains the number of logical address bits used for each table for table levels four through seven. The corresponding fields are ‘TIE’ to ’TIH’. The most significant bit of each field for ‘TIA’ through ‘TIH’ is stored in the fields labelled ‘A’ to ‘H’. In most cases the MSB will be zero.

	Extended Translation Control Register														
31	A	B	C	D	E	SREX		F	G	M		PSX	H		ISX_2	16
15	TIE					TIF			TIG				TIH			0

The ISX field adds two more bits to the IS field of the translation control register, allowing more of the upper address bits to be ignored during a translation.

Since the upper bits of a virtual address may indicate the read or write access level these bits will usually be ignored by the MMU.

PSX adds a bit onto the page size (PS) field allowing page sizes up to 2GB to be specified.

SREX adds a bit to the SRE bit forming a two-bit field. The two-bit field along with the operating mode indicates the root pointer selection.

The M field enables compatibility with the 68851. If M is set extra long format descriptors will not be recognized.

	{SREX,SRE}			
Operating Mode	0	1	2	3
0	CPU	CPU	CPU	CPU
1	CPU	SUPER	SUPER	SUPER
2	CPU	SUPER	HYPER	HPYER
3	CPU	SUPER	HYPER	SECURE

On reset the extended translation control register is set to zero.

Page

DT	Level 0	Level 1 to 7	Indirect Descriptor
Invalid 0	Invalid	Invalid	Invalid
Page 1	Type 1 page descriptor	Type 2 page descriptor	Type 1 page descriptor
Short 2	Indirect	Table descriptor	Illegal
Long 3	Indirect	Table descriptor	Illegal
Extra long 4	Indirect	Table descriptor	Illegal

Descriptor Formats

Existing table descriptor formats

For 64-bit physical addresses, the upper 32-bits come from the root pointer extension register. The short format table descriptor may not point to extra long descriptors.

<table>
<tr><th></th><th colspan="4">Short Format Table Descriptor</th><th></th></tr>
<tr><td>31</td><td colspan="4">$PA_{31\ldots16}$</td><td>16</td></tr>
<tr><td>15</td><td>$PA_{15\ldots4}$</td><td>U</td><td>WP</td><td>DT_2</td><td>0</td></tr>
</table>

The rf68851 uses bits in the long format table descriptor that were previously defined to be zero.

The long format table descriptor has an extra bit for the descriptor type allowing the specification of extra long format descriptors.

A table level field was added.

<table>
<tr><th></th><th colspan="9">Long Format Table Descriptor</th><th></th></tr>
<tr><td>63</td><td>L/U</td><td colspan="8">$Limit_{14\ldots0}$</td><td>48</td></tr>
<tr><td>47</td><td>RAL_3</td><td>WAL_3</td><td>SG</td><td>S</td><td>LVL_3</td><td>DT</td><td>U</td><td>WP</td><td>DT_2</td><td>32</td></tr>
<tr><td>31</td><td colspan="9">$PA_{31\ldots16}$</td><td>16</td></tr>
<tr><td>15</td><td colspan="6">$PA_{15\ldots4}$</td><td>~</td><td>~</td><td>$\sim_2$</td><td>0</td></tr>
</table>

Additional table descriptor format for rf68851

<table>
<tr><th></th><th colspan="9">Extra Long Format Table Descriptor</th><th></th></tr>
<tr><td>127</td><td>L/U</td><td colspan="8">$Limit_{14\ldots0}$</td><td>112</td></tr>
<tr><td>111</td><td>$RAL_{7\ldots5}$</td><td>$WAL_{7\ldots5}$</td><td>SG</td><td>S</td><td>LVL_3</td><td>DT</td><td>U</td><td>WP</td><td>DT_2</td><td>96</td></tr>
<tr><td>95</td><td colspan="9">$PA_{31\ldots16}$</td><td>80</td></tr>
<tr><td>79</td><td colspan="6">$PA_{15\ldots4}$</td><td>~</td><td>~</td><td>$\sim_2$</td><td>64</td></tr>
<tr><td>63</td><td colspan="9">$PA_{63\ldots48}$</td><td>48</td></tr>
<tr><td>47</td><td colspan="9">$PA_{47\ldots32}$</td><td>32</td></tr>
<tr><td>31</td><td colspan="2">$\sim_3$</td><td colspan="2">RGN_3</td><td colspan="2">$RAL_{4\ldots0}$</td><td colspan="3">$WAL_{4\ldots0}$</td><td>16</td></tr>
<tr><td>15</td><td colspan="9">$Limit_{30\ldots15}$</td><td>0</td></tr>
</table>

- The limit field has been extended to 31-bits allowing up to 2^{31} entries per table.
- RAL, WAL fields have been extended to eight bits allowing finer grained access levels.

- An additional 32-bit has been added to the physical address, physical addresses up to 64-bits may be specified.
- The RGN field is a three-bit index into an eight-entry region table describing associated memory regions.
- The LVL field reflects the table level that the descriptor points to.

Existing Page Descriptor Formats

For 64-bit physical addresses, the upper 32-bits come from the root pointer extension register.

	Short Page Descriptor (Type 1 & 2)								
31	$PA_{31\ldots16}$								16
15	$PA_{15\ldots8}$	G	CI	L	M	U	WP	DT_2	0

	Long Page Descriptor – Type 1											
63	~											48
47	RAL_3	WAL_3	SG	S	G	CI	L	M	U	WP	DT_2	32
31	$PA_{31\ldots16}$											16
15	$PA_{15\ldots4}$								~	~	$\sim_2$	0

	Long Page Descriptor – Type 2											
63	L/U	$Limit_{14\ldots0}$										48
47	RAL_3	WAL_3	SG	S	G	CI	L	M	U	WP	DT_2	32
31	$PA_{31\ldots16}$											16
15	$PA_{15\ldots4}$								~	~	$\sim_2$	0

Additional Page Descriptor Format (rf68851)

	Extra Long Page Descriptor											
127	L/U	$Limit_{14\ldots0}$										112
111	$RAL_{7\ldots5}$	$WAL_{7\ldots5}$	SG	S	G	CI	L	M	U	WP	DT_2	96
95	$PA_{31\ldots16}$											80
79	$PA_{15\ldots4}$								~	~	$\sim_2$	64
63	$PA_{63\ldots48}$											48
47	$PA_{37\ldots31}$											32
31	~	CL_2	RGN_3	$RAL_{4\ldots0}$	$WAL_{4\ldots0}$							16
15	$Limit_{30\ldots15}$											0

Existing Indirect Descriptors

	Short Indirect Descriptor		
31	$PA_{31\ldots16}$		16
15	$PA_{15\ldots2}$	DT_2	0

	Long Indirect Descriptor		
63	$\sim_{16}$		48
47	$\sim_{14}$	DT_2	32
31	$PA_{31\ldots16}$		16
15	$PA_{15\ldots2}$	$\sim_2$	0

Additional Extra Long Indirect Descriptor (rf68851)

	Extra Long Indirect Descriptor		
127	$\sim_{16}$		112
111	$\sim_{14}$	DT_3	96
96	$\sim_{16}$		80
79	$\sim_{16}$		64
63	$PA_{63\ldots48}$		48
47	$PA_{47\ldots32}$		32
31	$PA_{31\ldots16}$		16
15	$PA_{15\ldots2}$	$\sim_2$	0

Existing Invalid Descriptors

	Short Invalid Descriptor		
31	$\sim_{16}$		16
15	$\sim_{14}$	DT_2	0

	Long Invalid Descriptor		
63	$\sim_{16}$		48
47	$\sim_{14}$	DT_2	32
31	$\sim_{16}$		16
15	$\sim_{16}$		0

Additional Extra Long Invalid Descriptor (rf68851)

	Extra Long Invalid Descriptor		
127	$\sim_{16}$		112
111	$\sim_{14}$	DT_2	96
95	$\sim_{16}$		80
79	$\sim_{16}$		64
63	$\sim_{16}$		48
47	$\sim_{16}$		32
31	$\sim_{16}$		16
15	$\sim_{16}$		0

Access Level Registers

The access level registers are extended by five bits, allowing finer grain access. The upper three bits of the register retain their original purpose. When the MMU is operating in a compatible mode the lower five bits of the register are ignored.

CAL – Current Access Level

7 5	4 0
CAL	CALX

VAL – Valid Access Level

7 5	4 0
VAL	VALX